VUTLHARI KUBAYI

EXERCISE 2 - Aggregate Functions & Operators

Question 1

```
42  -- Q01 List distinct departments
43  SELECT DISTINCT department
44  FROM students;
45
46  -- Q02 Average age per department
```

Add parameter

Table

	department
1	IT
2	HR
3	Finance

Question 2

```
46  -- Q02 Average age per department
47  SELECT department, AVG(age) AS avg_age
48  FROM students
49  GROUP BY department;
```

Add parameter

Table

	department	avg_age
1	IT	20.5
2	HR	22
3	Finance	23

Question 3

```sql
-- Q03 Departments with more than 1 student
SELECT department, COUNT(*) AS student_count
FROM students
GROUP BY department
HAVING COUNT(*) > 1;
```

Add parameter

Table

	department	student_count
1	IT	2
2	HR	2

Question 4

```sql
-- Q04 Age BETWEEN 21 AND 23
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;
```

Add parameter

Table

	student_id	name	age	department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

Question 5

```sql
-- Q05 IT or HR and older than 21
SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT','HR') AND age > 21;
```

Add parameter

Table

	student_id	name	age	department
1	2	Bob	22	HR
2	5	Eve	22	HR

Question 6

```sql
-- Q06 Total credits per dept > 5
SELECT department, SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits) > 5;
```

Add parameter

Table

	department	total_credits
1	IT	11

Question 7

```sql
-- Q07 Courses not 4 credits
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits != 4;
```

Add parameter

Table

	course_id	course_name	department	credits
1	101	SQL Basics	IT	3
2	104	Excel	Finance	2
3	105	Statistics	HR	3

Question 8

```sql
-- Q08 Top 3 by credits
SELECT course_id, course_name, credits
FROM courses
ORDER BY credits
DESC LIMIT 3;
```

Add parameter

Table

	course_id	course_name	credits
1	102	Python	4
2	103	Data Science	4
3	101	SQL Basics	3

Question 9

```sql
-- Q09 Max, Min, Avg grade
SELECT MAX(grade) AS max_grade, MIN(grade) AS min_grade, AVG(grade) AS avg_grade
FROM enrollments;
```

Add parameter

Table

	max_grade	min_grade	avg_grade
1	90	78	84.6

Question 10

```sql
-- Q10 Count enrollments per course
SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

Add parameter

Table

	course_id	enrollment_count
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1

Question 11

```sql
-- Q11 Total salary and bonus per dept
SELECT department, SUM(salary) AS total_salary, SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;
```

Add parameter

Table

	department	total_salary	total_bonus
1	IT	122000	10500
2	HR	109000	7500
3	Finance	70000	6000

Question 12

```sql
-- Q12 Avg salary > 55000
SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

Add parameter

Table

	department	avg_salary
1	IT	61000
2	Finance	70000

Question 13

```sql
-- Q13 Salary + bonus > 60000
SELECT employee_id, name, salary, bonus, (salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60000;
```

Add parameter

Table

	employee_id	name	salary	bonus	total_compensation
1	1	Tom	60000	5000	65000
2	3	Spike	70000	6000	76000
3	4	Tyke	62000	5500	67500

Question 14

```sql
-- Q14 Total and avg budget per dept where avg > 70000
SELECT department, SUM(budget) AS total_budget, AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000;
```

Add parameter

Table

	department	total_budget	avg_budget
1	IT	270000	135000
2	Finance	80000	80000

Question 15

```sql
-- Q15 Budget BETWEEN 50000 AND 120000 excluding Marketing
SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000 AND department != 'Marketing';
```

Add parameter

Table

	project_id	project_name	department	budget
1	1	AI App	IT	120000
2	2	Payroll System	Finance	80000
3	5	HR Portal	HR	50000